

Multiple Choice (10 marks)

1. If the parabola $y_1 = x^2 + 2x + 7$ and the line $y_2 = 6x + b$ intersect at only one point, what is the value of b ?
 - (a) 7
 - (b) 3
 - (c) 12
 - (d) 4
2. Let $12-9+c=12$. What value of c makes the equation true?
 - (a) 0
 - (b) 3
 - (c) 9
 - (d) 12
3. What is a reasonable height for a doorway?
 - (a) 2 km
 - (b) 2 cm
 - (c) 2 m
 - (d) 2 mm
4. There are three real numbers x that are not in the domain of $f(x) = \frac{1}{1+\frac{1}{1+\frac{1}{x}}}$. What is the sum of those three numbers?
 - (a) 0.5
 - (b) 0
 - (c) -1
 - (d) -1.5
5. If $888x + 889y = 890$ and $891x + 892y = 893$, what is the value of $x - y$?
 - (a) 1
 - (b) -3
 - (c) -1
 - (d) 3

True / False (10 marks)

Answer True or False

6. True or False: The positive difference between the two solutions of the quadratic equation is 2.
7. True or False: 7% is equal to 0.07 when expressed as a decimal.
8. True or False: 26.1 mm is equal to 2.61 dm.
9. True or False: The value of a 4000 investment after 10 years, growing at the rate of $320e^{(0.08t)}$, is approximately 1000.
10. True or False: The two numbers that add up to 19 and multiply to 70 are 5 and 14.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Let a, b, c be the roots of $x^3 + 2x^2 + 5x + 7 = 0$. Compute

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

12. Discuss the relationship between the weight of an object on the moon and its weight on Earth. How can you express the moon's weight in terms of Earth's weight? Provide a rationale for your equation.

End of Paper